

Update Day 1 – January 27 2021

Location: Online (Pitch2Peer, BB, Academic Professional)

PPT-presentations (Videos)	
Haers	Ingeborg
Hardeman	Kevin
Heemskerk	Robin
Jansen	Demi
Karimi	Anahid
Brouwer	Ambroos
Maasland	Emma
Sieben	Naomi
Struik	Alieke
van Dijk	Yannick
van Mierlo	Shianne
Vossen	Bas
Weir	Alistair
Werner	Lea
Zantman	Cato

Poster-presentations (poster & written pitch)	
Beumer	Teun
Bogert	Dorien
Deng	NA
Gupta	Samiksha
Kalogeras	Dimitris
Makarov	Ivan
Mohammad	Fatemeh
Simons	Rogier
Teunisse	Martin
Tse	Ava
van der Gouwe	Tim
van Huissteden	Amber
van Leeuwenkamp	Alex
van Zijderveld	Remco
XIE	peiqi

PPT-Presentations

Name Student: Ingeborg Haers

Internship or Thesis: internship

Title project: increasing compliance to safety rules in the Efteling

Since the Efteling finds safety important, this study aimed to identify, categorize and gain an understanding of the most significant factors that affect compliance to safety rules.

Literature research revealed five categories that could affect compliance with safety rules:

(1) perceived organizational support, (2) perceived supervisor support, (3) safety motivation, (4) perceived ease of use, and (5) perceived usefulness. The results showed that perceived usefulness and perceived ease of use negatively affected compliance with safety rules. Two important factors within perceived usefulness and perceived ease of use were identified: clarity and attitude. Both factors could be increased by increasing the involvement of employees. Other factors were delay, flexibility of safety rules, cognitive workload, and ambiguity. Considering all these factors in the future will increase the perceived usefulness and perceived ease of use, meaning that compliance with safety rules will no longer be negatively affected by those categories.

Name company or University: Efteling

(If applicable) External supervisor: Frank Bersch

UU Supervisor: Susan te Pas

Name Student: Kevin Hardeman

Internship or Thesis: internship

Title project: Piloting a quantitative UX research method at Ruigrok NetPanel

In every statistics class, the idea being taught is that it is important to have a large enough sample size that is representative of the population you are testing. In qualitative User Experience research, like they are experts in at market research company Ruigrok NetPanel where I do my internship, this is not the case. The proven formula there is to test products and interfaces with 5 to 8 participants. I will talk about why in this case eight people can be enough and other things I learned at the university that do not always hold in the working place.

Name company or University: Ruigrok NetPanel

(If applicable) External supervisor: Anna Stekelenburg

UU Supervisor: Michiel Vellema

Name Student: Robin Heemskerk

Internship or Thesis: Internship

Title project: Development of a tool to examine non-verbal communication of uncertainty in clinical settings

This paper describes the development of a tool for examining non-verbal communication of uncertainty in an observational context based on video recordings of diagnostic consultations of Parkinson's disease. Such a tool does not yet exist, even though non-verbal communication significantly impacts patient satisfaction and the importance of uncertainty

management in healthcare has increased since the rise of concepts like evidence-based medicine, shared decision making, and patient-centered care. A mixed methodology was used, starting with the development of an observation table based on a general overview of the literature. This observation table was used by the researchteam to qualitatively observe the non-verbal expressions of the clinician during verbal expressions of uncertainty. These observations were discussed within the researchteam and the most promising non-verbal cues were identified to be further investigated. If sufficient basis was found in the literature for it as a non-verbal sign of uncertainty, it was added to the final product, which was a coding instrument for non-verbal communication of uncertainty.

Name company or University: Amsterdam UMC

External supervisor: Elisabeth Kurpershoek

UU Supervisor: Baptist Liefoghe

Name Student: Demi Jansen

Internship or Thesis: Internship

Title project: Dealing with cognitive biases in Service Design processes

Zeewaardig Service Design uses design methods and skills to help clients solve complex social challenges such as citizen participation, energy transition or innovating educational program. They facilitate (digital) workshops and design sprints where they, together with clients and actors, work towards creative solutions. During my internship I looked into how Zeewaardig can apply cognitive psychology in the company. I created two card decks which they can use to address cognitive biases in different parts of the design proces. One deck focused on how to reduce biases during qualitative analysis and the other deck focused on how to reduce biases during social situations such as brainstorming or group decision-making. Next to this I gave a two-hour internal workshop to the whole company where I gave an introduction into relevant ACP theories and taught them how to use to card decks in their own work and when facilitating workshops.

Name company or University: Zeewaardig Service Design

External supervisor: Adinda de Lange

UU Supervisor: Maarten van der Smagt

Name Student: Anahid Karimi

Internship or Thesis: Internal Internship

Title project: Gaining situational awareness in semi-automated driving: a predictive model.

Objective: In this study I present a mathematical model of takeover readiness in semi-autonomous driving, by simulating several different ways of gaining situational awareness.

Background: As the driving industry gradually shifts from manual to (semi-)automated vehicles, more research on safety risks and human-machine interaction is needed. Previous studies proposed frameworks for the stages of the transfer process and the factors that play a role. However, less attention focused on how situational awareness after disengagement from the road is gathered and what the influence is on takeover readiness. **Method:** The model used here provides an explorative prediction of what a safe deadline for presenting an external alert for take-over might be. The model tests the impact of different mathematical (linear, exponential, diminishing returns) functions on takeover readiness.

Application: The model integrates data from over 60 previous studies and can contribute to the design of safe and effective semi-automated vehicles.

Keywords: semi-autonomous driving, SAE level 3, situational awareness, transfer of control, human-automation interaction, driver distraction.

Name company or University: Utrecht University

UU Supervisor: dr. Chris P. Janssen

Name Student: Ambroos Brouwer

Internship or Thesis: Internship

Title project: Mouse-tracking to investigate dynamics of cognitive processes: A systematic review of current data-analysis methods

Mouse-tracking can be a useful tool to uncover the cognitive processes underlying a participant's motor response during decision-making or categorization tasks. Originally, mouse coordinates are recorded from which simple measures are calculated in order to determine the amount of decisional conflict that occurs as a result of competing choice options. These measures include maximum deviation and area under the curve. While these measures have shown to be effective in summarizing simple trajectory dynamics, their simplicity becomes problematic when trying to summarize trajectories with higher dynamical variance. Moreover, both maximum deviation and area under the curve only measure a trajectory's spatial features, while disregarding its temporal characteristics. Multiple researchers have therefore recently developed new methods for the analysis of mouse-tracking data. The goal of this systematic review is to provide an overview of these methods, and to determine for which applications they may or may not be useful. While these new approaches have been shown to be useful for reasoning research, analysis of new trajectory characteristics or improved trajectory visualizations, they are not necessarily more effective at measuring decisional conflict than the conventional method.

Name company or University: UU

UU Supervisor: Chao Zhang

Name Student: Emma Maasland

Internship or Thesis: Internal Internship

Title project: Developing a programming assignment in Python for the bachelor Psychology

During my internship, the goal was to write an assignment about functions in Python for students of the bachelor's degree in Psychology. This assignment was designed to be part of a new course called 'Programmeren'. Through this process, my personal goal was to learn how to program using Python. The process that led to this assignment can be split up into three distinct phases: First came a learning phase, where I had to teach myself all necessary Python skills. The second phase was an iterative design process where I received continuous feedback from my supervisors. The third and final phase was the prototyping phase, where the assignment was tested by people from different backgrounds and skill-levels concerning Python. The results are one assignment, which has two versions (teachers' and students'), and a package of supplementary code. Additionally, I can state that I have fulfilled my personal learning goal for this internship: learning Python.

Name company or University: Utrecht University

UU supervisor: Roy Hessels

Name Student: Naomi Sieben

Internship or Thesis: Thesis

Title project: Noise and Visual perception of Urban Air Mobility vehicles

Currently there is a lot of popularity for the future use of Urban Air Mobility (UAM) vehicles, such as drones and air taxis. It is expected that these vehicles will be used in the future for parcel delivery and even passenger transport. A negative aspect of this implementation is noise pollution. During my internship at the NLR I will measure noise annoyance from drones using a Virtual Community Noise Simulator (VCNS). Through virtual reality glasses and noise cancelling headphones, a drone is either visually presented with the corresponding sound, or only the sound is presented. A questionnaire is then filled in to measure the level of annoyance elicited by these vehicles.

Name company or University: Nationaal Lucht- en Ruimtevaartlaboratorium (NLR)

External supervisor: Roalt Aalmoes

UU Supervisor: Jana Klaus

Name Student: Alieke Struik

Internship or Thesis: Internship

Title project: Division of labour in cognitive collaboration

During my internship at Lund University, I did an experiment about cognitive collaboration. Dalmaijer et al. (2018) investigated if people would change their behaviour during cognitive collaboration when there is a difference between their own and their collaborators capacity. They set up an experiment with a visual short-term memory task. People first completed the task individually to determine their short term memory capacity. After that, they worked with another participant on the task. Together they needed to remember eight items, and each person could claim one or more items to remember. The conclusion of their study was that people did not act upon the differences in capacity. This study however, had a few limitations. My internship goal was to complete another experiment to finish the study. I carried out an experiment where people did the same tasks, but this time they worked together with a computer. More emphasizes was put towards the scores and after each round they needed to estimate their partners capacity. Participants divided labour equally and not in proportion with capacity differences.

Name company or University: Lund University

External supervisor: Diederick Niehorster

UU Supervisor: Martijn Mulder

Name Student: Yannick van Dijk

Internship or Thesis: Internship

Title project: A journey from the laboratory to the practical world. How knowledge translation can make scientific research more engaging for a broader audience.

Recently, a paper by GEMHlab (games for emotional and mental health lab) was published in the Journal of Medical Internet Research (JMIR). The JMIR has asked GEMHlab to translate this paper into a more engaging format that is suitable for a broader audience. Several examples of usable knowledge translation formats were discussed and examined. These include video, 'printed' materials like infographics and comic books, and interactive elements incorporated into the paper. Eventually, the choice was made to focus on a comic book as a potential format. A mock-up was created and given to a small panel of youth advisors. Their feedback helped shape the current version of the scientific comic book. The comic will be further developed and finalised, after which it is ready to be published as an addition to the article to serve as a format that makes science easier to understand, more accessible, and more engaging for a younger audience.

Name company or University: GEMHlab

External supervisor: Nastasia Griffioen

UU Supervisor: Anouk Keizer

Name Student: Shianne van Mierlo

Internship or Thesis: Internship

Title project: Distinctive Brand Assets: Benchmarks and development over time

Distinctive Brand Assets (DBA's) are all the aspects, apart from the brand name, that trigger a certain brand in someone's memory. DBA's are measured on fame and uniqueness. However, there are different types of DBA's, i.e. colour, human face, shape, sound, visual and words, that score differently. Therefore, to judge whether a specific type of DBA performs well or not, benchmarks were created using data from old and new online surveys. The results showed that all benchmarks except for Visual – Component scored below 50% on fame, while there are a few subtypes that scored more than 50% on uniqueness. In addition to the benchmarks, it is important to know how DBA's develop over time. The results showed a lot of variation in change over time scores. In addition, there are too many unknown variables that influence fame and uniqueness scores to truly draw a conclusion.

Name company or University: Validators

External supervisor: Marceline Veen

UU Supervisor: Zsafia Belteki

Name Student: Bas Vossen

Internship or Thesis: Internship

Title project: Grapedistrict Newsletter 2.0

Grapedistrict is an off- and online wineshop in Amsterdam. Existing customers are subscribed to the newsletter of Grapedistrict. At Grapedistrict, the newsletter is a key moment to introduce customers to new wines, offers and campaigns. By using various tools, the mail is made as personal and direct as possible. The effectiveness of the newsletter is measured by three aspects: open-rate, click-rate and conversion. A/B-tests are used to determine which tools have the biggest impact. Examples of tools are use of colour, manner of addressing and visual cues. These tools are tested separately and combined to look for existing correlations.

Name company or University: Grapedistrict

External supervisor: Myrthe Roelofsen

UU Supervisor: Leendert van Maanen

Name Student: Alistair Weir

Internship or Thesis: Internship

Title project: An 'Expert Review' on a Patient Bedside Monitor's User Interface

This paper evaluates the user interface of a medical bed-side patient monitor. The portal is used by healthcare professionals (HCP) to monitor patient vital signs during MRI scans. The objectives of this study were to identify use-related issues the user may experience while operating the system and provide specific recommendations for any opportunities for improvement. The findings conclude that while there are positives to the device's user interface overall, there are many opportunities for improvement. Applications for future research and studies include human-computer interaction and user interface design.

Name of company: Emergo by UL

External supervisor: Tamara Will

UU Supervisor: Dr Krista Overvliet

Name Student: Lea Werner

Internship or Thesis: Internship

Title project: The Mouse-Tracking Project

For my Master's in Applied Cognitive Psychology at Utrecht University, I pursued my internship at Alpha.One, a neuromarketing company based in Rotterdam that is specialized in the use of neuroscientific research methods. The aim of the project I worked on aimed to explore the utility of using a mouse-tracking task as an alternative for eye-tracking in the context of webpage saliency. We developed a study in which the participants were first presented a series of webpages meanwhile their eyes were tracked; three weeks later we repeated the same webpage presentation implemented in a mouse-tracking task. We then generated the heatmaps from the registered fixations in order to subsequently examine their similarities. We do not have final results yet, however, high similarity of the heatmaps

would suggest that the research methodology under investigation represents an attainable alternative to eye-tracking data collection.

Keywords: visual attention, webpage saliency, eye-tracking, mouse-tracking

Name company or University: Alpha.One

External supervisor: Coert Van Gemeren

UU Supervisor: Baptist Liefoghe

Name student: Cato Zantman

Internship or thesis: internship

Title project: What are you looking at? - A manual for and evaluation of using Unity as a tool for research into gaze behavior

This manual is intended to function as a guide for researchers that plan on doing research into gaze behavior using virtual characters in Unity. Three packages are discussed to implement and model different aspects of humanlike gaze behavior: Inverse Kinematics can be used to make the head and eyes of a character follow an object; the Realistic Eye Movements package will enable the user to create humanlike gaze behavior in a prefabricated character; the Heretic package has an extremely detailed rig laying over the face of the character with which the facial expressions of the character can be adjusted. Recommendations for future developments are: a situation in which all discussed techniques and packages can be combined, more complex environments to test the effects of different settings in gaze behavior, and humanoid characters with independently manipulable 'bones' for the eyes and the eyelids accessible to the public.

Name company or University: Utrecht University

UU supervisor: dr. Roy Hessels

Poster-presentations

Name Student: Teun Beumer

Internship or Thesis: Internship

Title project: Creating a dashboard with R shiny

The goal of this internship is to create an online dashboard for real life data-visualisation with R shiny, a function of R studio. During the first weeks of the SARS-Covid-19 pandemic the Utrecht University launched a wide-scaled study to monitor the changing of habits during the first 10 waves of the pandemic. The Utrecht University monitored the changing of habits such as the automatization of wearing a face mask in a supermarket, or keeping distance with strangers in a supermarket. Now it is time to make the data accessible to the public by the means of a dashboard in which it will be able to plot several variables. In the process, skills such as programming in R and how to host a small website were learnt.

Name company or University: University Utrecht

UU Supervisor: dr. Chao Zhang

Name Student: Dorien Bogert

Internship or Thesis: Internship

Title project: Yesterday's bread for tomorrow - Advice report on gauging interest among Dutch bakers regarding sustainable bread making

Food production and consumption have a large, negative environmental impact, especially food waste. *Bakkersgrondstof* reduces food waste amongst Dutch bakeries. They use a special machine that makes new raw material in the form of fermented bread flour from old, unsold bread. As the project *Bakkersgrondstof* is still at the early stages of development, the project can still take many turns. This report is dedicated to the question what the best way is to gauge interest among Dutch bakeries in joining this project. A nationwide questionnaire was send out by the *Nederlandse Brood- en banketbakkers Ondernemersvereniging* (NBOV), gauging the interest amongst artisanal bakeries. Yet, the number of responses ($n = 2$) was too low to analyze. This report discusses the original research process, its limitations that might have led to this result, and what steps *Bakkersgrondstof* can take to really reach bakeries in the future. The main advice is yet to be formulated.

Name company or University: HubOranje!

External supervisor: Albert Hoekerswever (albert@highopes.nl)

UU Supervisor: Sjoerd Stuit (S.M.Stuit@uu.nl)

Name Student: Na Deng

Internship or Thesis: internship

Title project: prepare a scientific research experiment.

Firstly, I searched the literature related to my research topic, and found that most of the current research of synthetic speech on emotion have not classified the scale of pitch in the fine branch. The aim of my experiment is to understand the influence of synthetic speech on

the attribution of affect to synthetic voices. this study will test the differences on a finer-grained scale using five different Pitch contours and Rates. In the progress of my internship, I use sets of semantically unpredictable sentences as the text of the speech. I used Google's text-to-speech to convert the text sentence into the synthesized speech, and then edited those speech with Adobe Audition. Furthermore, I decided to use the robot scale of pick-a-mood and SAM scale and got the permit access from original author by email. Finally, I used E-prime software to edit the experimental procedure.

Name company or University: Eindhoven university of technology (TU/E)

External supervisor: Dr Peter Ruijten-Dodoiu

UU Supervisor: Dr. Stella Donker

Name: Samiksha Gupta

Internship or thesis: internship

Title: Patient Engagement in MRI: Towards a better User Experience

MRI scans often are associated with feelings of anxiety and claustrophobia. The patient's stress and anxiety during the Magnetic resonance Imaging (MRI) scan often leads to poor image quality due to patient movement, as well as incomplete scans which leads to a delayed diagnosis. In certain cases, an extremely unpleasant experience can cause prolonged psychological effects such as fear of enclosed spaces and loud noises. The present study, in the form of literature overview, aims to obtain insights into patient's anxiety and experiences during MRI and Radiotherapy (RT), and which type of intervention and techniques can be most relevant to alleviate patient's discomfort during the treatment, thereby providing a pleasant experience to an otherwise stressful situation.

Name of Company: Philips

UU Supervisor: David Terburg

External Supervisor: Sanne Nauts

Name Student: Dimitrios Kalogeras

Internship or thesis: Internship

Title project: Implementations of Nonverbal Behavior in Games: Art and Psychology

This project inquires how game animators implement nonverbal behaviors, especially facial expressions, and gaze, in their characters. As sculptors, animators can intuitively produce acting characters. The goal of this paper is to inquire if animators make use of principles based on psychology research. Games were assessed on a basis of "what are the implementation techniques?" and "what are the principles chosen to satisfy their games' goals?". Findings show that both games for research and commercial games utilize the categorization of behaviors in repertoires, tied to dialogue segments. This addresses the multimodal nature of nonverbal behavior. Animators are also informed from psychology literature on facial expressions. Nonetheless, animators choose principles depending on their goals, which would be to transfer the essence of a scene in an efficient processing cost. Animators would recreate and emphatically depict behaviors, which shows that a readable behavior may be more important than a realistic one.

Keywords: video games, nonverbal behavior, facial expressions, gaze

Name Company or University: Utrecht University

External supervisor:

UU Supervisor: Roy Hessels

Name Student: Ivan Makarov

Internship or Thesis: Combined Internship

Title project: Exploring the interaction between the pupillary light response after exogenous attentional cueing and detection performance

The pupil changes size as a response to light. Previously, it was generally assumed that the pupillary light response (PLR) is modulated only by luminance, while recent studies have shown that the PLR is also modulated by cognitive processes, for example, by the allocation of spatial attention. This attentional modulation of the PLR has been hypothesized to facilitate detection and discrimination of visual information. The goal of the current research was to investigate whether the PLR influences perception. The PLR was evoked by cueing effect in that appears in the different parts of the screen (white and black). Perception was operationalized as a detection performance for threshold stimuli. Results showed that when a pupil is dilated, the threshold stimuli detection when they appeared on a dark side of the screen. In conclusion, results of the current study are in line with previous findings that a decrease of the pupil size improves detectability, while constriction enhances discriminability of small stimuli.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Stefan Van der Stigchel

Name Student: Fatemeh Mohammad

Internship or Thesis: Internship

Title project: Attachment in workplace: the role of sense of equity and community

Aim: The goal of this study is to describe the role of equity and community in the workplace to increase the sense of commitment and attachment to the employing organisation and also to identify the psychological, environmental, and architectural factors that enhance the sense of equity and community in the professional environments. **Findings:** The conducted review formulates the determinant socio-psychological factors involved in the perception of the sense of equity and community in organizations, using the Kotter and Heskett's framework (1991) for organizational factors. It also shows how the workplace could hit the middle ground in providing sufficient spaces for contradictory but essential needs of employees, such as the need of adequate space for communication and on the other the need for privacy, or the need of feeling the equity in the organization and on the other hand the unavoidable existence of hierarchy which should be fine-tuned to enhance the sense of fairness among the employees. **Study type:** Literature review

Name company or University: MMEK

External supervisor: Erik van Kuijk (erik@mmek.nl) and Martijn Meeske (martijn@mmek.nl)
UU Supervisor: UU 1st supervisor: Stella Donker, UU 2nd supervisor: Chris Janssen

Name Student: Rogier Simons

Internship or Thesis: Internship

Title project: Automatic Object Detection for Video Annotation

This project is part of a combined Internship & Thesis on Mobile Eye Tracking research. A big problem with Mobile Eye Tracking research is that its data analysis is a very time-consuming task. For the thesis we want to investigate whether it is possible to decrease analysis time by using automatic object detection, and if so, by how much? But first, we must delve deep into the world of Automatic Object Detection to familiarize ourselves with these techniques.

Name company or University: University

(If applicable) External supervisor:

UU Supervisor: Maarten vd Smagt

Name Student: Martin Teunisse

Internship or Thesis: Internship

Title project: Creating course material for online research

In this project I started of doing researching different kind of tools that could be used for making experiments online. After the research I selected 5 programs for testing. Jatos, Limesurvey, Pavlovia, Qualtrics and Gorilla. I tested each of them on the amount of tools the programs possess, usability, data safety and available support tools. Jatos was quite easy to learn how to build experiments although not easy to set up servers for the project to work on. Limesurvey had excellent servers already set up to be used but the program is quite hard to learn. Pavlovia might be the future of online research they have a very good offline program called PsychoPy in which they are continuing new updates. From that program you are just one click away until the experiment is online. At the moment the program still has a few bugs and beta functionalities. Not everything that will work offline will work online. But I am confident one day it will be very good. There were two excellent programs existing each with their own merit. Qualtrics and Gorilla. Qualtrics is an excellent tool for building questionnaires and making them look good. In Gorilla however you can make Questionnaires as well as other kinds of experiments. Due to this Gorilla is the overall best program to be used. It has servers set up for you. It has a great support page. Loads of tools to be creative with and there are loads of example projects for how to use them.

Name company or University: University Utrecht

UU Supervisor: Roy Hessels

Name Student: Ava Tse

Internship: Internship

Title Project: Can people categorize emotions from color and movement in a robot?
Designing non-verbal behavior with a simple spherical robot

This project aims to provide an extensible approach on designing emotionally expressive behavior in low-degree of freedom and constrained appearance robots. Can people categorize emotion from color and movement from a robot without limbs or head? An empirical study is carried out on a commercially available simple spherical robot, Sphero. The robot is designed to present happy and sad emotion by applying two psychological models: Laban Movement Analysis for movement design and Plutchik's Wheel of Emotions for LED color. 78 participants attributed scores based on video clips with the different combination of LED colors and movement paths in Sphero. Experimental results show movement with applying Laban's model gives significant clue for emotion classification, however, LED color with Plutchik's theory is not an effective tool in emotion expression.

Name company or University: Utrecht University

UU supervisor: Roy Hessels

Name Student: Tim van der Gouwe

Internship or Thesis: Internship

Title project: How bad UX leads to fewer donations

The importance of good user experience has been widely discussed in scientific literature. It has shown to improve conversion rates and overall customer satisfaction. Due to COVID-19 it is of paramount importance that customers can work with a website in an efficient, effective and satisfactory manner. By conducting a remote usability study we identified multiple usability issues on the website of Stichting Hoormij. Participants said that over half of the issues identified stopped them from finding the donations page. In short, Stichting Hoormij's website contains several usability issues that are likely to have a negative impact on the donation

Name company or University: User Sense

UU Supervisor: Jeroen Benjamins

Name Student: Amber van Huissteden

Internship or Thesis: Internship

Title project: Sequential process in decision-making and the effect on pupil size

The aim of the internship was to design and conduct a pilot experiment. The goal was to test a relatively new framework for the effects of decision-making on pupil dilation (Strauch, Koniakowsky & Huckauf, 2020). The framework assumes a sequential process including an initial binary decision on the goal-relevance of a given stimulus and a subsequent further evaluation only of relevant stimuli. When stimuli are considered goal relevant, the pupil dilates. Around 200 ms later, a further evaluation of relevant stimuli takes place with pupil

dilating stronger for relatively higher evaluated stimuli. The research question that we wanted to answer during the internship is 'how are effects of differential consequences affecting pupil dilation and possibly reaction times?' to test the framework. Participants needed to complete a sequential letter matching task. The consequence to errors were manipulated. An eyetracker was used to measure the pupil size. At this moment, data collection is still going on. There are no final results to show yet. Nonetheless, first data analysis shows cautious support for the framework. The final findings in this pilot study can be used for further research.

Name company or University: University Utrecht

UU Supervisor: Christoph Strauch

Name Student: Alexandra van Leeuwenkamp

Internship or Thesis: Internship

Title project: The Future of the Workplace

Recent attention given to the importance of commitment and engagement by corporation management has heavily centered on attachment and employee emotional well-being. Organisations understand the need to create a workforce that feels attached to their company, the benefits of this have shown to have a large impact on productivity, greater innovative behaviour, and well-being. A company's ability to create a clear meaning and identity expressing their underlying beliefs and values will not only help establish attachment with their employees but also outside individuals. The goal of this paper is to highlight the current scientific understanding of how workplace identity and meaning are essential to help encourage a higher level of attachment. Furthermore, I will establish how the built environment of the workplace is an important aspect in which identity and meaning can be delivered. There are three key elements in the built environment that have been established as important to manifest identity and meaning in the workplace, these are, corporate communication, corporate design, and corporate culture.

Name company or University: MMEK Design for Experience, Utrecht

External supervisor: Martijn Meeske and Erik van Kuij

UU Supervisor: Dr. Chris Janssen

Name Student: Remco van Zijderveld

Internship or Thesis: Internship

Title project: The Paced Auditory Serial Addition Task (PASAT) with monitoring feedback in Python: programming a stress task for PTSD patients.

The main goal of the internship was to create a stress task from the Python environment with feedback from a physiology monitoring system called BIOPAC. The task that was created was the Paced Auditory Serial Addition Task (PASAT) with feedback on performance and the monitoring of physiological responses. This neuropsychological task is already used in measuring the working memory. By providing negative feedback to this task, the PASAT becomes a rather stressful task. Due to delivering problems from the manufacturer, the inclusion of BIOPAC is unfortunately still missing. To compensate for this inconvenience,

feedback is provided based on correct response streak. In a broader sense, I've gained extra skills in programming in the python language. Besides that, I have a better understanding of physiological responses. The PASAT is going to be used for my thesis which will focus on body regulation of PTSD patients during stressful circumstances.

Name company: Brain Research and Innovation Centre (BRAIC), Military Mental Health

External supervisor: Antoin de Weijer

UU Supervisor: Ruud Hortensius

Name Student: Peiqi Xie

Internship or Thesis: Internship

Title project: Investigating the Performance of Mid-Air Input in Virtual Reality: Comparison of Hand-Up vs. Hand-Down Gestures

Despite the interaction technology for virtual reality (VR) systems has evolved significantly over the past years, the text input efficiency in the virtual environment is still an ongoing problem. This study aimed to investigate the performance of the word-gesture text entry technology with different input postures and VR experiences in the virtual environment. The study revealed that the VR experience had little effect on input performance, while the hand-up posture had a positive influence on input performance with word-gesture text entry technology in a virtual environment. In addition, the typing accuracy and user experience for using the hand-up posture were higher than that for the hand-down posture. The hand-up posture also had less task workload than the hand-down posture. This study proves that the word-gesture text entry technology with hand-up posture has greater application potential than hand-down posture.

Name company or University: Tsinghua University

External supervisor: Jibo He

UU supervisor: Stella F. Donker